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Predicting and forecasting disasters: A global scan of traditional and local knowledge

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ABSTRACT

For centuries, people worldwide have predicted disasters based on observations and experimentation, interpreting animal behavior, plant responses, weather patterns, and celestial phenomena. Despite its significance, such traditional and local knowledge remains under-researched and undocumented, limiting its potential for integration in disaster risk reduction. This review consolidates traditional and local knowledge from 53 research articles, covering 423 cases categorized into three broad indicators: astronomical, meteorological, and biological. These indicators encompass 33 sub-indicators, such as sky, wind, birds, and soil, providing location- and disaster-specific contexts for prediction. Meteorological disasters (e.g., tropical cyclones, storms, and mass movements) constituted the largest share (33 %), followed by hydrological (e.g., floods and storm surges) and climatological disasters (e.g., droughts) (27 %). While there were disaster-specific variations, animal behavior (mammals, insects, birds, etc.) were the most commonly used predictive indicator (39 %), followed by water-related indicators (12 %), plant phenology (9 %), wind (8 %), and both cloud patterns and temperature (5 % each). Other indicators, including observations of the sun, moon, sky, stars, lightning, and rainfall, collectively constituted the remaining 22 %. There were notable similarities and differences in disaster prediction within and across countries in terms of the indicators used. It is, therefore, important to contextualize and localize prediction patterns rather than generalize them. However, scientific metrics need to be explored to assess their broader applicability. This would be a crucial step in harnessing traditional knowledge for integrating effective prediction methods, which requires increased funding and research efforts.

1. Introduction

Humans have long faced various disasters, ranging from earthquakes, floods, and landslides to forest fires [1]. Disasters are rare and catastrophic events that significantly impact human well-being, leading to loss of life, injuries, property damage, and the destruction of infrastructure [2]. Many disasters occur naturally, such as earthquakes, volcanic eruptions, storms, floods, and wildfires, while others,

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including war, pollution, nuclear incidents, and transportation accidents, are human-induced [3]. The extent of damage to humans depends on their level of exposure and the resilience of the affected community [4]. Over the past two decades (2000–2019), 7,348 disasters triggered by a natural hazards have been recorded worldwide, claiming 1.23 million lives, affecting 4 billion people, and resulting in economic losses totaling US$2.97 trillion [5]. While the devastating impacts of natural disasters cannot be entirely prevented, their consequences can be mitigated through early warning system, preparedness and risk reduction strategies [6].

Forecasting and predicting disasters are fundamental to disaster risk reduction (DRR) [7]. These processes represent the initial stages in the DRR warning system, reducing vulnerability and mortality through effective preparation [3,8]. Advances in monitoring and forecasting methods, such as satellite imagery, surveillance systems, hydrological and meteorological modelling, and artificial intelligence, have significantly enhanced disaster prediction capabilities in recent decades [9,10]. Despite technological progress, vulnerability to natural hazards continues to rise globally, affecting both developed and developing nations [9]. Some modern technologies are costly and unavailable to many who might benefit [11]. Many communities do not have access to these technologies due to limited internet availability, communication infrastructure and awareness [12,13]. People around world have been using clue to predict and forecast disasters. Such clues need to be documented and interpretated in the given context to strengthen traditional knowledge-base, which, in some case, can be instrumental to develop a cost-effective and efficient prediction system [14]. Therefore, it is important to acknowledge that existing technological advancements alone are insufficient, necessitating integrating local and traditional knowledge for effective disaster management [9].

Traditional and local knowledge (TLK), often referred to as traditional knowledge (TK), traditional ecological knowledge (TEK), or local ecological knowledge (LEK), is invaluable for disaster prediction and preparedness [15]. Communities in hazard-prone regions acquire this knowledge through experience, intuition, and teachings, enabling them to interpret natural indicators such as animal behavior, plant conditions, geological changes, constellations, meteorological phenomena, and climatic patterns [6,16]. This knowledge helps identify hazard signals, develop adaptive strategies, and effectively communicate risks [9].

The importance of TLK in disaster management has been recognized since the 1970s and gained further prominence after the 2004 Indian Ocean earthquake and tsunami [17–19]. Global frameworks, including the Sendai Framework for Disaster Risk Reduction 2015–2030, the Global Framework for Climate Services, and the Sustainable Development Goals (SDGs), underscore the need of integrating local knowledge into DRR strategies [20]. The Sendai Framework emphasizes leveraging local knowledge for disaster risk assessment and the development and implementation of early warning systems. The United Nations Environmental Program (UNEP) recognizes the role of local and traditional knowledge in the management of natural disasters [11]. Despite being recognized in policies and plans [19,21–23], the TLK for disaster prediction remains underutilized due to documentation gaps and the preference for modern technological solutions [24,25].

Studies have shown scientific merit of TLK based on animal behavior and natural phenomenon, in predicting and forecasting disaster such as earthquakes, tsunamis, typhoons, and floods [16,26,27]. For instance, elephants can detect low-frequency vibrations associated with seismic activity, while insects respond to changes in magnetic fields and atmospheric gases [28]. Plants have been observed to exhibit structural changes before extreme weather events, further demonstrating the predictive potential of ecological indicators [6]. However, not all traditional predictions are scientifically validated, and some remain controversial. The objective of this study is not to authenticate every aspect of TLK but to systematically catalog documented local knowledge in disaster prediction and forecasting. Here, we discuss some TLK with possible scientific validation based on available literature to offer insight into how seemingly unlikely indicators and phenomena can have scientific merit. Such discussion also supports new research and innovations in disaster prediction based on various indicators.

This study aims to provide a consolidated overview of the global application of TLK in disaster prediction based on empirical studies to date, and to explore their underlying scientific basis where available. It seeks to contribute to the integration of TLK into DRR strategies and to empower communities to better withstand and cope with the impacts of natural hazards.

2. Materials and method

2.1. Data sources and search strategy

We searched research papers in both Scopus and Web of Knowledge, two of the largest scientific databases for research in social and environmental sciences. We used search strings combining any of one of the knowledge systems and disasters to find original research articles on traditional and local knowledge (TLK) in disaster risk reduction. Accordingly, we used either one of following strings in both databases to search them in title, abstract and keywords: "traditional knowledge*", "local knowledge*", "indigenous knowledge*", "co-production of knowledge*". The same approach was used to find papers containing one of the following disasters: disaster*, landslide*, tsunami*, earthquake*, volcano*, storm*, drought*, avalanche*, fire*, wave*, epidemic*, "insect infestation*", "animal stampede*".

The search resulted a total of 1,577 articles from Scopus and 1,206 articles from Web of Knowledge. After removing duplicates, the remaining articles (n = 1688) underwent a two-stage screening process. The first screening involved a review of titles and abstracts to identify studies relevant to the role of TLK in forecasting and predicting disasters, which resulted in the exclusion of 1502 articles. The remaining 186 articles were subsequently re-screened, where the full texts of the articles were evaluated against predefined exclusion criteria. The exclusion criteria included: (i) secondary studies, systematic reviews, or unsupported publication types (e.g., editorials, short notes etc.); (ii) articles with poor methodologies (e.g., those that do not clearly explain how data were collected), inaccessible content, or written in languages other than English; (iii) studies focused on man-made disasters such as traffic accidents, or chemical, biological, nuclear, and explosive events; and (iv) articles not focusing on TLK in predicting or forecasting disasters. Following the rescreening, 53 articles were identified as eligible, which were considered for data extraction and synthesis.

2.2. Data collection process

Once the text search was completed, data were extracted from the full text of the selected articles and organized using a pre-designed data grid. The extracted variables included the country of study, the type of disaster, and the details of the prediction method. To classify disasters, we adopted the framework by Ref. [29] which categorizes disasters based on whether they are natural or human-induced and further divides natural disasters based on their origin into five main groups: biological, geophysical, hydrological, climatological, and extra-terrestrial events. These broad groups were further subdivided into main types, sub-types, and sub-subtypes to achieve a more detailed categorization of disaster events. We used sub-type classification, which includes avalanche, rock fall, landslide, subsidence, earthquakes, tsunamis and volcanic eruption under the geophysical category; tropical cyclones/storms, extra-tropical cyclones, local/convective storms under meteorological category; and generic (river) floods, flash floods, storm surge/coastal floods under the hydrological category. While landslides, valances and subsidence are also included in the hydrological category, we considered them in geophysical category due to ambiguity regarding their origin. Not all of the subtype listed above were covered in this review.

Prediction data related to disasters were coded based on the elements involved in the prediction (e.g., objects, events, etc.) to identify similarities in indices used. The selection of indices was refined through successive iterations. Finally, the indices were organized into three primary indicators and 28 sub-indicators to have a structured and detailed analysis of TLK in disaster prediction and forecasting (Table 1, Fig. 1).

2.3. Data analysis

We analyzed reported cases of disaster prediction by country and continent. We computed the frequency distribution of disaster instances by main types and sub-types, as well as the distribution of various indicators and sub-indicators. We cross-matched the indices used for different types of disasters across countries and visually represented them using Sankey diagrams. We then discussed the primary indicators for each disaster sub-type and presented the corresponding sub-indicators in the tabular form. All indicators were examined in detail based on their potential scientific rationale, as reported in peer-reviewed scholarly literature available at Scopus and Web of Knowledge databases. Our aim here is not to validate all reported indicators, but rather to provide a synopsis of their possible scientific basis. This also demonstrates how seemingly unrelated indicators may, in fact, have a scientific explanation when examined closely.

3. Results

Of the 53 papers analyzed, 412 cases were identified in predicting and forecasting using TLK. The number of cases per paper ranged between 1 and 40 with an average of 7.92.

3.1. Spatial distribution of disaster prediction and forecasting cases

The cases documented TLK from 25 countries (Fig. 2), with the Philippines having the highest count of prediction/forecasting cases (n = 55), followed closely by Kenya (n = 48). There was an average of 16.4 disaster cases per country. Most of the cases reported were from Asia (46 %), followed by Africa (39 %), Oceania (11 %) and Europe (3 %). Australia contributed just 1 % of total cases. This indicates paucity of similar researches in North and South America.

3.2. Categories of reported disasters and their prediction proxies

The prediction and forecasting cases differed significantly across disaster groups ($\chi^2 = 506.95$, df = 4, $p < 0.000$), with meteorological disasters accounting for one-third of the documented cases. Within this category, cyclones (56 %) and typhoons (35 %) constituted the majority, while generic storms comprised only 9 % of the total. Hydrological disasters represented the second most reported cases (31 %), with river floods accounting for half of these, followed by flash floods (25 %). Climatological disasters constituted 28 % of the cases, predominantly represented by droughts (96 %). Geophysical disasters accounted for only 7 % of the total cases, primarily consisting of earthquakes (48 %) and volcanic eruptions (38 %) (Fig. 3).

A total of 28 indicators were identified as being used to predict and forecast disasters, but there were marked differences. The most

Table 1
Indicators of disaster prediction and forecasting through TLK.

Primary indictors	Explanation
Biological	Biological indicators encompass physical, morphological, phenological, and social traits of plants and animals that exhibit observable responses to environmental conditions, enabling local communities to identify early warning signs of potential disasters [30].
Meteorological	Meteorological indicators relate to phenomena such as clouds, rain, temperature, water (e. g., rise/fall water level), wind, thunder, and lightning, which provide valuable insights into weather patterns, including drought [31]
Astronomical	Astronomical indicators are related to celestial bodies and phenomena [32]. They include a description of physical appearance of celestial objects such as the moon, sun, and stars based on their size, shape and color in relation with other factors in given time and location.

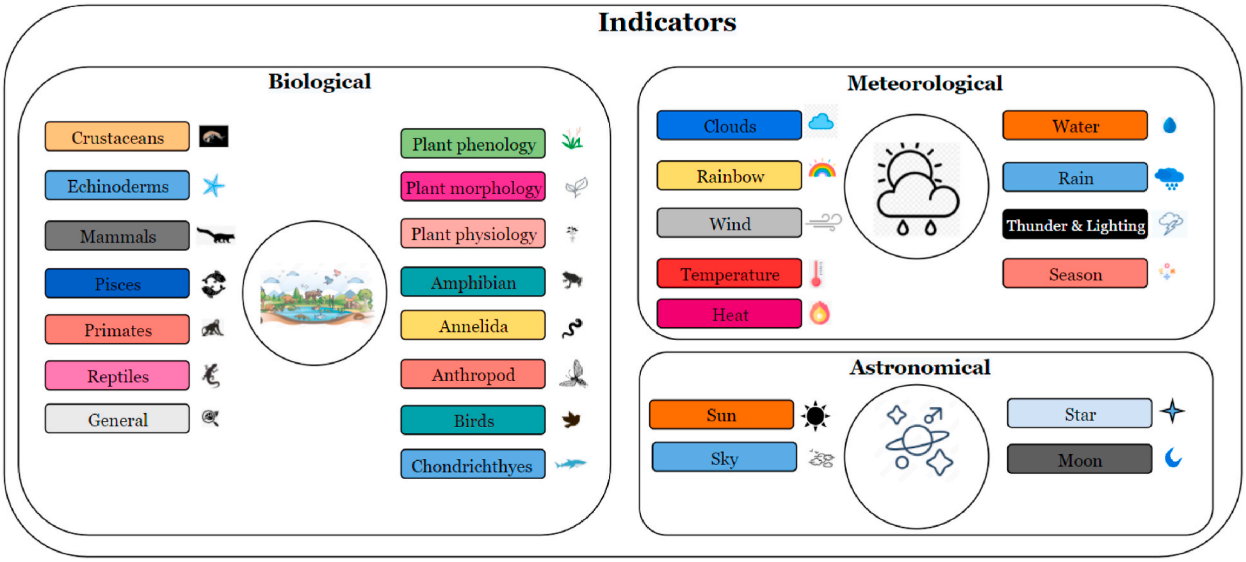

Fig. 1. Classification of primary indicators and sub-indicators that provide a basis of predicting and forecasting disasters by local communities.

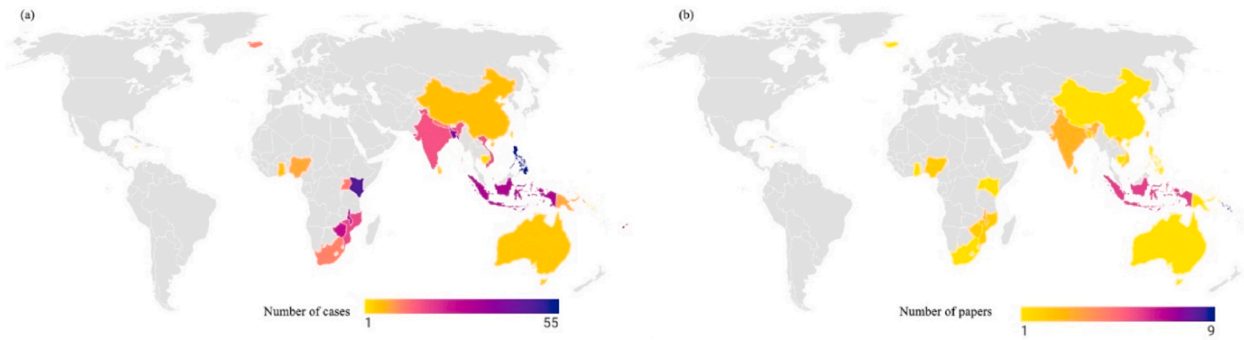

Fig. 2. Spatial distribution of disaster cases: (a) number of disaster cases and (b) number of publications by countries.

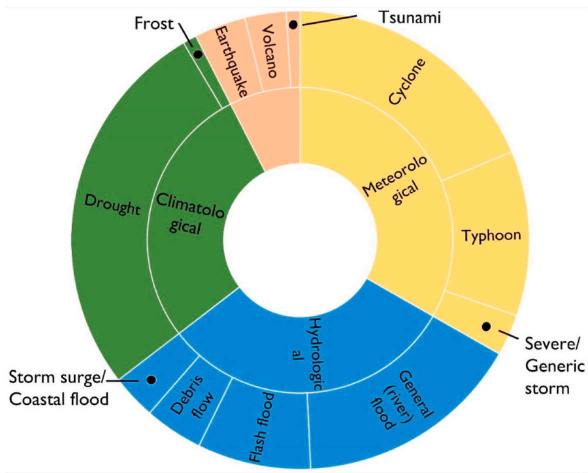

Fig. 3. The number of prediction and forecasting cases by disaster group and sub-groups.

common indicators were plants (15 %), followed by water (13 %), birds (12 %), mammals (9 %), wind (8 %), temperature (5 %), and clouds (5 %) (Fig. 4), suggesting that biological indicators are widely used in disaster prediction. However, there were variations in the use of indicators depending on the type of disaster. Droughts are most often predicted by the behavior of living organisms (e.g., insects, birds and mammals), including plant phenology (e.g., emergence of leaves and flowers, etc.). Astronomical and meteorological phenomena are reported as proxies for predicting disasters (Fig. 5). While earthquakes and volcanic eruptions are predicted using several proxies, the behavior of birds, mammals, and reptiles was most frequently reported. Hydrological disasters exhibit a diverse pattern of indicators, ranging from plant phenology to water surface changes. Floods (general river flooding) showed the most diverse proxies (Fig. 5), largely due to their widespread occurrence worldwide. Similarly, meteorological disasters—cyclones and typhoons—are predicted through atmospheric changes and the behavior of birds, fish, and crustaceans, alongside celestial signs.

3.3. *Similarities in prediction among countries and continents*

There were notable similarities in the use of sub-indicators for predicting and forecasting disasters across different countries. Cyclones were predicted using bird behavior in Australia, Bangladesh, and Fiji. Similarly, astronomical indicators, such as the position of the moon and stars, are used in Kenya and Mozambique in Africa to predict droughts. Meteorological factors, such temperature, wind, and water surface conditions, played a significant role in flood and typhoon predictions in the Philippines and Vietnam (Fig. 6). In Malawi and South Africa, floods are predicted based on arthropod behavior. Additionally, plant phenology and animal behavior, including that of mammals and insects, are widely used across different contexts.

3.4. *Mechanism of predicting/forecasting climatological disasters*

Climatological disasters are extreme weather or climate events that occur due to long-term changes in weather patterns, usually driven by broader climate conditions over an extended periods (months to years) [33]. These disasters are often linked to shifts in global or regional climates, such as droughts or heatwaves [34].

3.4.1. *Drought*

Drought is a situation when rainfall is deficient, below average for a long period of time [35]. It can last for weeks, months, or even years. Drought impacts are often slow to develop, but the consequences can be long-lasting and severe [36].

3.4.1.1. Astronomical indicators. Table 2 provides some circumstances within the annual seasonal calendar where the appearance of celestial objects may provide a contextual rationale for drought prediction. “Appearance of star”, “many star” and “bright star at night” are listed as indicators of impending drought [37–39]. Such conditions, where a clearly visible star appears for an extended period, may indicate a lack of clouds in the sky (Table 2). Reduced cloud cover exacerbates surface heating, soil moisture loss, and atmospheric moisture deficits, all of which intensify drought conditions [40,41].

The moon plays a significant role in weather prediction, with its phases, color, position, shape, and the size of its halo being key indicators used to forecast drought in various regions. Halo formations around the moon, caused by the refraction of light through ice crystals in cirrus clouds, often signify moisture in the upper atmosphere, which can lead to precipitation [42] (Table 2). However, halos

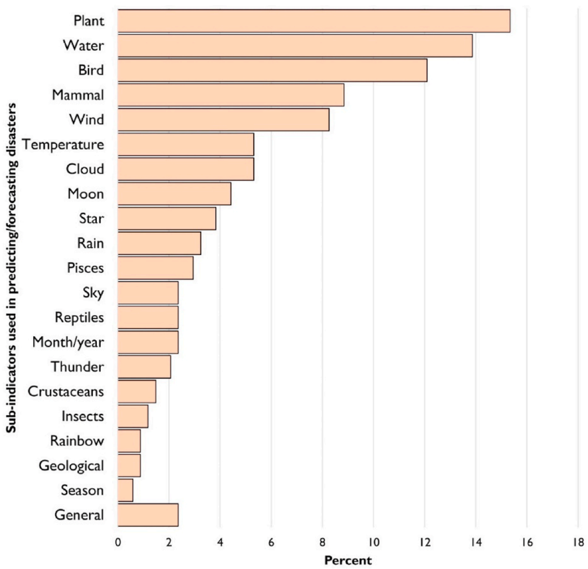

Fig. 4. Sub-indicators used in predicting/forecasting disasters.

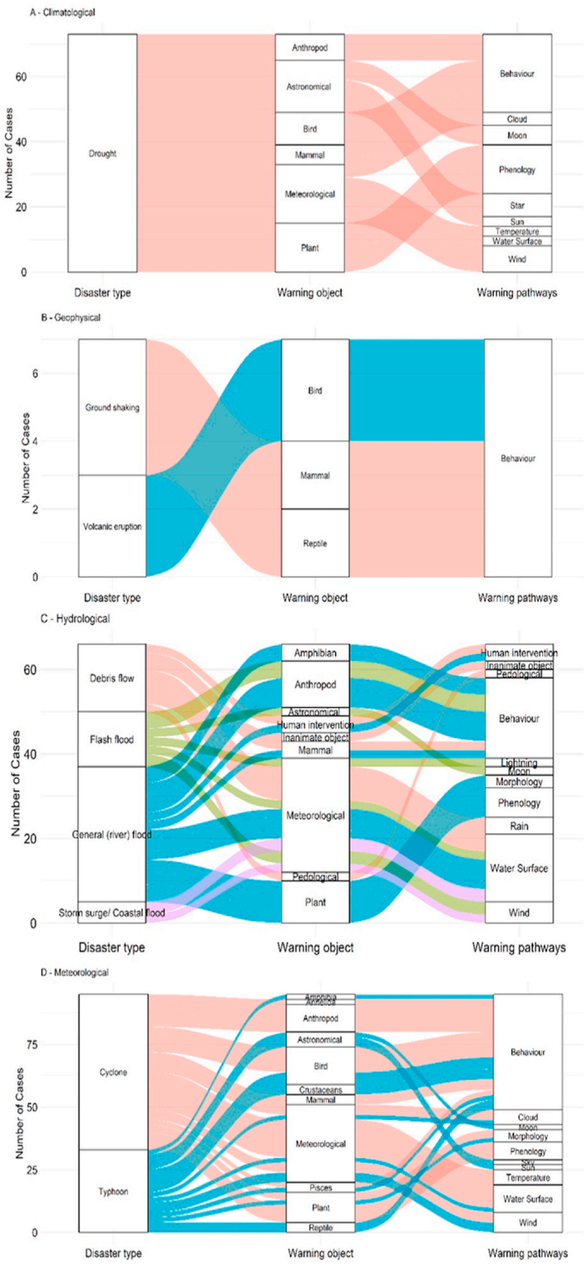

Fig. 5. Sankey diagram showing linkage between disasters, indicators and sub-indicators. Only cases greater than and/or equal to five are shown.

observed without subsequent cloud development may indicate insufficient moisture to result in rain, suggesting a potential lack of precipitation [43]. While some studies suggest a weak, non-significant correlation between moon phases and rainfall patterns over long periods [44], some other noted extreme rainfall and its temporal variations corresponding to the specific lunar phases, such as waxing gibbous, waning gibbous, and waning crescent [45]. In Kenya, the full moon phase is linked to drier periods, suggesting a relationship between lunar cycles and weather patterns [46]. Observations of the influence of the Earth's revolution around the Sun on cloud formation, rainfall mechanisms, and seasonal cycles are discussed in early intellectual works, such as the *Upanishads* from 3000 BCE [47]. The position, shading, and presence or absence of a halo around the Sun have been used to predict drought [37,39]. The attributes of the Sun and Moon can enhance our understanding of and ability to predict drought and rainfall variability in data-deficient regions.

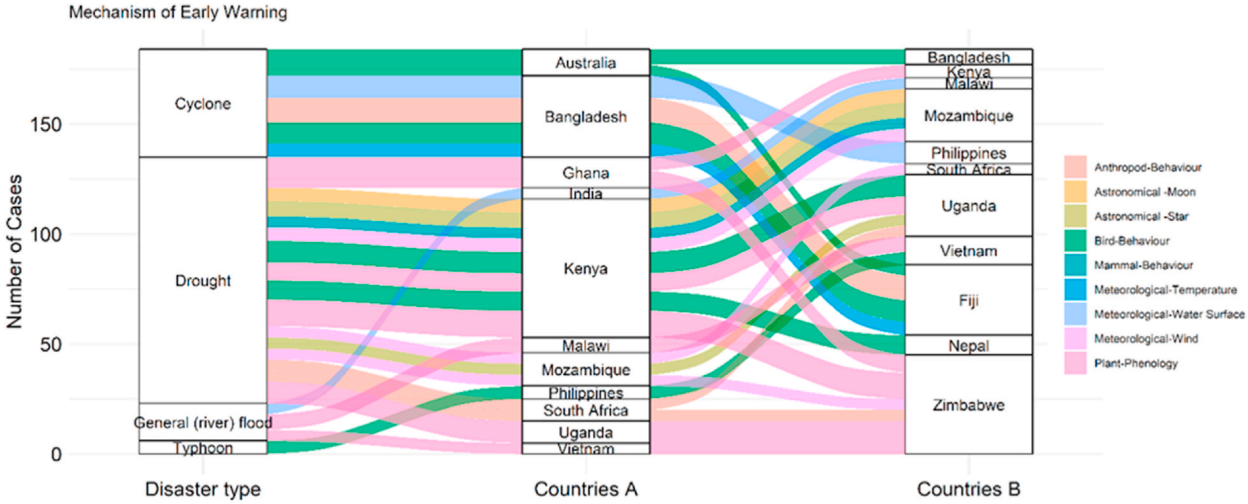

Fig. 6. Similarities in disaster prediction in terms of indicators.

Table 2
Astronomical indicators used in TLK for predicting and forecasting droughts.

Moon	In Kenya, a visible full moon signifies a dry period, while a dark moon phase indicates a wet period. Additionally, the shape and size of the moon, as well as the absence of a ring or clouds around it at night, are associated with drought [39,46]. In Zimbabwe, a larger halo or ring around the moon suggests a wet period, whereas a smaller halo or its absence indicates a dry period (Mutasa, 2015). In India, the color of the moon with a pale-yellow hue predicts good rainfall, while red or white tones indicate dry conditions [48]. In Mozambique, the position of the full moon is observed: if it rises perpendicularly or inclined, normal conditions are expected. However, if it rises "upside-down," appearing lower on the horizon with its illuminated side facing away from the earth, it may signal changes. Additionally, the absence of a ring or clouds around the moon at night is associated with drought [37,39].
Star	In Uganda, a night sky full of stars indicates that the morning will be clear, without clouds or rainfall [38]. In Mozambique, clear stars without cloud shadows, stars that appear to move and brighten the earth, and stars dispersed across the sky are used as indicators. Similarly, a sky full of numerous stars also observed in both Mozambique and Kenya to signify specific weather patterns [37,39]. In Kenya, the appearance of a constellation of seven stars in the east signals drought, while its central position indicates a non-drought season as the position of stars in the night sky are different in different seasons [39].
Sun	In Mozambique, a clearly visible sun without clouds or a circle around it is used as an indicator [37]. In Kenya, the presence of a very small circle around the sun and the position of the sun—specifically when it passes to the left of the Capricorn—are considered significant in predicting drought [39]. When the Sun enters Swati (one of the 27 Nakshatras (lunar mansions) in Hindu astrology), it causes some occasional rain otherwise the south-west monsoon withdraws totally [47].

3.4.1.2. Biological indicators. Table 3 provides short narratives about biological phenomena and how people worldwide relate them to changes in environmental conditions, often using them as indicators of drought [30,49,50]. Here, unusual behaviors (e. g., movements or vocalizations, etc.) of living organisms such as frogs and snakes during such periods are taken to indicate impending drought. For example, the selection of specific microhabitats (e. g., sites with abundant rocks and dense bush cover) by *Tropidosaura wiegmanni* (a lizard) during the summer, despite the availability of other habitats, is perceived by local people as unusual and indicative of a drought to come [51].

Insects are the most prominent group and their morphology, behaviour and social organization are used as indicators of drought. For example, marching ants carrying eggs [50], yellowish maggots with black stripes, unified locust movements [52], insect outbreaks (e.g., ants, grasshoppers) [53], bees flying toward northern hills (Sarkar et al., 2015), and low density of spider webs [54] indicate drought in different parts of world (Table 3). Ants, bees, and butterflies often migrate or exhibit collective behaviors, such as locusts moving in a unified direction or colonies of ants carrying eggs, which local people use to predict and forecast drought (Table 3). These behaviors may result from vapor pressure deficits in the air at specific locations, serving as indicators of approaching drought [55]. Similarly, drought stress in plants can alter the chemical signals they emit, affecting the behavior of herbivorous arthropods. For instance, drought-stressed plants may release volatile organic compounds that influence herbivores, potentially changing their feeding behavior, movement, and reproductive strategies [56,57]. Behavioral adaptations, such as increased activity in search of moisture, have also been documented in various terrestrial arthropods, underscoring their capacity to respond to immediate environmental stressors [55,58].

Birds provide valuable clues about changing environmental conditions, with their responses closely observed by local communities through continuous monitoring and experience [59]. This includes migration pattern and behaviors. For example, Thari birds migrating west to Mbeere [60], Kamakumetsi birds flying in large groups over long distances [38], hens spreading feathers under balconies [61], flocking of birds [62], unusual bird sounds and movements [63], disappearance of weaver birds and directional migrations indicate drought in different part of world (Table 3). Such unusual behaviors, specific to a location, are a reflection of drought-induced circumstances. For instance, drought often leads to a decline in vegetation cover, reducing habitat quality and prompting many avian species to migrate to more favorable environments [49,64,65]. In semiarid regions, alterations in rainfall

Table 3
Biological indicators used in TLK for predicting and forecasting droughts.

Amphibia	Frogs vocalizing in the afternoon in South Africa [52]
Arthropoda	In Ghana, a marching colony of ants carrying eggs [50] and the appearance of small yellowish maggots with black stripes signal changes in rainfall, with the maggots indicating an upcoming drought, and their movement into the earth suggesting imminent rainfall. Similarly, the appearance of certain ants, such as Muthua and Nduti, also points to drought, while the presence of dragonflies, butterflies, and other insects signifies non-drought conditions. In South Africa, an army of locusts moving in a unified direction may suggest drought, while a kaleidoscope of butterflies flying together [52] indicates favorable weather. The movement of bees, such as those flying toward northern hills in India [48], signals dry periods, while their movement toward southern hills indicates good rainfall. In Zimbabwe, the low density of spider webs [54] is linked to drought, while a higher density indicates wet conditions. Similarly, in Uganda, the appearance of flying and crawling insects [62] could be a sign of shifting weather patterns. In Kenya, bees moving from east to west [39] may signal changes in environmental conditions. Additionally, outbreaks of insects like ants and grasshoppers in South Africa [53] could indicate drought or environmental stress. In livestock, fewer births and abortion in cattle, as well as a decline in mopane moths and worms in Zimbabwe [70], are also signs of impending drought.
Birds	In Kenya, flocks of Thari birds migrating from the west to Mbeere, as well as their tendency to invade maize crops [60], suggest environmental stress, possibly linked to drought. Similarly, unusual bird behavior such as the Kamakumetsi birds in Uganda [38] flying in large groups of over 2,000 individuals across distances of 50–100 km, indicates changes in the weather. In Nepal, hens spreading their feathers under the balcony [61] may reflect potential drought. In Uganda, the flocking of various bird species [62] points to shifts in weather patterns, while unusual sounds and movements of birds in Kenya [63] further suggest potential environmental stress. The appearance of rare birds, like the Kaunga and sparrow weaver, often signals drought, while the appearance of many birds in large groups, such as Nthungululu and Nzonzwe, typically occurs in non-drought periods. The disappearance of certain birds, like weaver birds, or the migration of birds from west to east or south to north, can also indicate shifts in weather, while weak poultry [39] may reflect poor environmental conditions pointing to drought.
Mammals	In South Africa, horses playing joyfully and the burning skin of rabbits [52] are notable signs of unusual environmental stress. In Mozambique, animals displaying unusual behavior, such as not running or playing as usual [37], signal possible disruptions in their environment. Similarly, in Kenya, the constant weakness of poultry, cattle, and humans, along with domestic animals like poultry and cattle fighting over food [39], indicate scarce resources likely caused by drought. The invasion of wild animals, such as zebras, into human settlements typically points to drought conditions, while the appearance of numerous wild animals like Mbaa, squirrels, and Nganga birds attacking germinating crops suggests a period of non-drought. In Kenya, the birth of many goats, especially twins, is considered a good sign, indicating favorable conditions [60].
Reptiles	The movement of snakes in the same direction, as observed in South Africa [52], may serve as an indicator of environmental changes, such as shifts in temperature or the approach of a drought
General	The natural death of wild animals, usually between August and October [Zimbabwe] [74]. Black intestine of slaughtered animals [Kenya] [63].
Plant Morphology	Seaweed is seen in an upright position [Philippines] [69]. Khair trees (*Acacia catechu*) become extra bushy [India] [48]. Leaves of Maulu, Kiwua Nzuki, Yiulu trees turning yellow or shedding [Kenya] [39].
Plant Phenology	Flowering of Dawa dawa plant [Ghana] [50]. Shedding of leaves of tree species such as Kukhuyu (Ficus mucuso) and Kuchihiri (Cordia africana) [Uganda] [38]. High abundance of wild fruits prior to the farming season [Zimbabwe] [54]. Flowering of Muvau, Kinguthi, and Maulu trees [Kenya] [39]. Ever green and too many leaves blooming of Umevha (*Acacia*). Less fruit in Umtopi (*Boscia albitrunca*) in september–October and less fruits in Umhagawuwe (*Securinega virosa*) in November–March. Too many flowers and green leaves of Uphane (*Colophospermum mopane*) (September–December). Too many leaves and fruits in Xakuxaku (*Azanza garckeane*) [Zimbabwe] [70]. Flowering of wild plants. Ripening of fruits and shedding of leaves of Milicia excelsa [Uganda] [62]. Trees shedding their leaves [Kenya] [63]. Flowering of trees e.g. *Acacia* sp., Kithumula, Mumbu, Baobab, Kithiia, Mataa Mwaka, Kiatine, Kiusi, Kiua, Nyumbililya. Flowers in Mataa Mwaka tree indicates end of dry season and if the flowers wither before the onset of rains, then it will indicate insufficient rainfall. No flower or late blooming in trees like Acacia sp., Muthiia, Mwa and too much bloomed in trees like Acacia, Muthiia, Maulu, Baobab. Ripening of Thwaala, unusual shedding of the flowers of the mango tree and flowering of Muthiia tree [Kenya] [39].

patterns can significantly impact bird abundance and community structure, making migratory birds particularly vulnerable to drought [64]. More intense droughts are consistently associated with fewer birds occupying affected areas [66]. Birds may exhibit migratory restlessness in response to unfavorable conditions, a behavior that serves as an internal cue for migration [67] which is perceived as an indicator of drought by local people.

Mammals and domestic animals exhibit various behavioral changes in response to environmental stressors, including weakness, hunger, and unusual activities. Examples include horses playing joyfully, rabbits experiencing a burning sensation on their skin [52], and animals displaying atypical behaviors such as reduced movement or playfulness [37]. Additionally, poultry, cattle, and humans may show signs of weakness, while domestic animals may fight over food due to scarcity [39] (Table 3). These behaviors are linked to physiological mechanisms that detect changes in blood volume, pressure, and osmotic balance, signaling the central nervous system to initiate thirst and conserve water [68]. As a result, animals may exhibit signs of weakness, engage in unusual activities, or compete for limited resources.

Plants exhibit sophisticated signals to detect and respond to drought conditions, often before these effects become apparent to humans (Table 3). Various plant species show distinct morphological and phenological changes. For example, seaweed stands upright [69], and Khair trees (*Acacia catechu*) become extra bushy [48]. Leaves of Maulu, Kiwua Nzuki, and Yiulu trees turn yellow or shed [33], while the Dawa dawa plant begins flowering [50]. Tree species like Kukhuyu (*Ficus mucuso*) and Kuchihiri (*Cordia africana*) shed their leaves [38].

Other signs include a high abundance of wild fruits before the farming season [54] and the flowering of Muvau, Kinguthi, and Maulu trees. Evergreen and abundant leaf blooming occur in Umevha (*Acacia*), while less fruit is observed in Umtopi (*Boscia albitrunca*) between September and October and in Umhagawuwe (*Securinega virosa*) from November to March [39]. Excessive flowering and green leaves appear in Uphane (*Colophospermum mopane*) from September to December. Additionally, Xakuxaku (*Azanza garckeana*) shows abundant leaves and fruits [70]. Wild plant flowering, fruit ripening, and leaf shedding in *Milicia excelsa* are also reported as drought indicators by local communities (Table 3) [62]. Here, a key hormone in this process may be abscisic acid (ABA), which regulates plant responses to abiotic stresses, including drought [71]. During water deficits, increased ABA levels cause physiological changes such as stomatal closure, reducing water loss [72]. This leads to observable phenological changes in plants. The root system plays a critical role in sensing soil moisture levels. Hydraulic signals from the roots communicate with the shoot system, triggering systemic responses like stomatal closure [73].

3.4.1.3. Meteorological indicators. Various meteorological indicators such as rainfall, cloud, thunder and lightning are used to predict droughts. Studies showed that some of the indicators include frequent cloudy skies without rainfall [50], disappearance of fog by 7 a.m. or extended cloudless periods during the rainy season [37], clear skies or dark cumulus/nimbus clouds appearing only during the day [39] and a faint yellow sky, black-colored clouds [47] (Table 4).

Cloud formations, such as overcast skies and specific cloud types like cumulus or nimbus, often signify drought or the potential onset of rain (Table 4). For example, a strong correlation (0.84) exists between deep convection cloud (DCC) frequency and precipitation, showing that reduced cloud cover is closely associated with decreased rainfall [75]. The absence of clouds contributes to increased solar radiation and evaporation, intensifying dry conditions [76]. Cumulus clouds suggest atmospheric instability and potential precipitation, but their dissipation without rain may signal an approaching dry period. Conversely, nimbus clouds are often associated with rainfall, but if they fail to produce precipitation, this could indicate a shift toward drier conditions.

Other meteorological factors, such as rainfall delays, extreme heat or cold, and unusual temperature fluctuations, often serve as drought signals. Delayed or low rainfall at the start of the farming season [39] is reported as an indicator of drought, as it directly affects soil moisture and agricultural productivity [77]. This correlation is particularly evident in regions where agricultural practices are heavily dependent on timely precipitation, a typical case in rain-fed areas. Temperature-related indicators such as excessive heat from August to October [70], a sudden and consistent rise in surface temperatures [62], persistent extreme heat year-round [37,63] and unusually sunny and hot weather, or unusual coldness and low temperatures [39] can also signify drought (Table 4). Excessive heat, especially during critical growing periods, has been associated with increased drought severity [78,79]. A sudden and consistent rise in surface temperatures can exacerbate moisture deficits, leading to hydrological drought as surface water resources diminish [80]. Furthermore, persistent extreme heat year-round can significantly impact vegetation health and water availability, further compounding drought conditions [79]. Unusual weather patterns, such as prolonged sunny and hot conditions or unexpected cold spells, can disrupt normal climatic cycles, thereby serving as indicators of impending drought [77].

Water-related indicators such as absence of frost from June to July [70], morning mist [39,62] and drying up of water bodies [39] are signals of drought for communities in different parts of world (Table 4). The absence of frost can indicate higher temperatures that may lead to increased evapotranspiration, thereby reducing soil moisture levels [81]. Similarly, morning mist can contribute to local humidity levels, which can mitigate drought effects by providing moisture to plants during critical growth phases [77]. The drying up of water bodies is another significant indicator, as it reflects the cumulative effects of reduced precipitation and increased evaporation rates, signaling severe drought conditions [77].

The mechanisms behind these observations are rooted in the interactions between meteorological variables and hydrological responses. For example, the propagation of meteorological drought to hydrological drought is influenced by factors such as precipitation deficits and temperature anomalies, which can lead to prolonged periods of low water availability [79]. Meterological drought occurs when there is a prolonged period of below-average precipitation (rainfall, snow, etc.). It is the earliest stage of drought and is defined based on climate conditions, such as reduced rainfall over weeks, months, or years. Hydrological drought occurs when the lack of precipitation (meteorological drought) leads to reduced water levels in rivers, lakes, reservoirs, and groundwater sources. It takes longer to develop because it depends on how precipitation deficits impact water bodies over time. Studies have shown that meteorological droughts often precede hydrological droughts, with significant time lags observed in various regions [80]. This lag can be

Table 4
Meteorological indicators used in TLK for predicting and forecasting droughts.

Cloud	Frequent cloudy skies with no rainfall, where the sun rises around 1 p.m. after a cloudy morning and the sky becomes overcast again by 4 p.m. until nightfall for approximately three weeks, signify an impending drought in Ghana [50]. In Mozambique, the disappearance of fog by 7 a.m. or an extended period without clouds, even during the rainy season, indicates drought [37]. In Kenya, a very clear sky or the appearance of dark cumulus or nimbus clouds during the day are associated with drought [39]. A faint yellow sky or crow-colored clouds during the day with a clear night sky suggests low chances of rain. Rainfall accompanied by sunshine is also considered an indicator of a lack of rain in the near future [47].
Rain	A prolonged delay in rainfall or low rainfall at the start of the farming season is considered an indicator of drought in Kenya [39].
Temperature	Excessive heat from August to October is an indicator of drought in Zimbabwe [70]. In Uganda, a sudden and consistent rise in surface temperature signifies drought [62]. Persistent extreme heat throughout the year is associated with drought in Mozambique and Kenya [37,63]. Additionally, unusually sunny and hot weather or unusual coldness and low temperatures are also considered signs of drought in Kenya [39].
Water	The absence of frost (Ungqwaqwani) from June to July is considered an indicator of drought in Zimbabwe [70]. The appearance of morning mist is associated with drought in Uganda and Kenya [39,62]. Additionally, the drying up of water bodies is a sign of drought in Kenya [39].
Wind	Southerly winds from September to October are considered an indicator of drought in Zimbabwe [70]. In Nepal, winds blowing from southeast to northwest during the monsoon season signify potential drought [61]. In Mozambique, drought is associated with warm, fast-moving winds, winds blowing consistently from one direction, or winds blowing from two opposite directions, such as west and east [37]. In Kenya, strong winds or back-and-forth winds from Mount Kilimanjaro before the rains are also seen as drought indicators [39].
Thunder and Lightning	Lightning from only one direction, as observed in Mozambique [37], and frequent thunderstorms accompanied by rare lightning, as documented in Kenya [39], are indicators of drought conditions.

attributed to the time required for reduced precipitation to translate into lower streamflow and groundwater levels, highlighting the interconnectedness of these drought types.

Wind patterns are important indices, such as southerly winds from September to October [70], winds blowing southeast to northwest during the monsoon [61], warm, fast-moving winds, consistent unidirectional winds, or winds from opposite directions at the same time (west/east) [37], strong winds or erratic winds near Mount Kilimanjaro before rains [39], suggest drought conditions or onset of drought (Table 4). Prevailing winds from arid regions frequently carry dry air masses, reducing humidity and increasing evaporation. This phenomenon is particularly evident in regions influenced by the El Niño-Southern Oscillation (ENSO), where shifts in wind patterns significantly impact local climates [82]. ENSO-driven wind changes also influence seasonal drought predictability globally. For instance, extreme drought events in Pakistan strongly correlate with wind patterns, highlighting the role of intrinsic weather systems in drought forecasting [83]. Similarly, wind speed and other meteorological factors are critical for predicting drought conditions in Bangladesh [84]. Prolonged low wind speeds are often linked to drought, particularly in the context of climate change [85]. Changes in sea surface temperatures can alter wind directions, affecting rainfall distribution. This relationship is essential for understanding seasonal weather patterns and predicting extreme weather events, including droughts and floods [82].

Thunder and lightning are also indicators of drought such as lightning appearing from only one direction [37] and frequent thunderstorms accompanied by rare lightning [39]. These phenomena highlight how specific patterns of thunder and lightning are used by local communities to predict environmental stress.

3.5. *Geophysical disasters*

Geophysical disasters are natural events originating from processes and phenomena within the Earth's crust, mantle, and core. These events often have significant impacts on human populations, ecosystems, and infrastructure. They are typically caused by tectonic activity, volcanic activity, or gravitational forces. They include earthquakes and volcanic eruptions.

3.5.1. *Earthquake*

An earthquake is the sudden shaking or vibration of the Earth's surface caused by the release of energy from within the Earth's crust [86]. This energy is usually the result of tectonic movements, volcanic activity, or other geological processes. People use different signals from astronomy, animals, plants, insects and meteorological evidence to forecast earthquakes.

3.5.1.1. Biological indicators. Biological indicators of earthquakes encompass a range of phenomena observed in both animal behavior and physiological changes (Table 5). These indicators can serve as potential precursors to seismic events, reflecting the complex interplay between environmental stressors and biological responses.

Some of the reported indicators include frogs, insects, fish, lizards and worms behaving unusually [87], insects like crickets intermittently stopping sounds for two-three minutes during the night [88], unusual behaviors in cats, dogs, cows, mice, elephants, horses, and pigs [53], dogs barking frequently, looking at the sky, crawling on the ground, and howling [89], bats flying away abruptly, snakes coming out of their holes and huge flocks of white herons returning to the mainland during the daytime [90] (Table 5). One of the most notable biological indicators is the unusual behavior exhibited by animals prior to earthquakes. For instance, studies have documented that common toads (*Bufo bufo*) exhibited significant behavioral changes, such as leaving breeding sites days before the L'Aquila earthquake in Italy in 2009 [91,92]. This behavioral response has been attributed to the animals' ability to detect subtle changes in their environment, which may include shifts in chemical composition of groundwater that precede seismic activity [91,92]. Such changes can trigger stress responses in animals, leading to alterations in their typical behaviors, which may serve as indicators of impending earthquakes (Table 5).

Annelids, like worms, exhibit atypical activities, while arthropods, including ants, crickets, and other insects, show disruptions in their normal patterns. Arthropods, particularly insects, are known for their acute sensitivity to environmental changes. Research indicates that certain species exhibit altered behaviors prior to seismic events, potentially due to their ability to sense vibrations and changes in the Earth's magnetic field [93]. For instance, some studies suggest that insects may detect the low-frequency sounds or vibrations generated by tectonic movements, which precede an earthquake [91]. This heightened sensitivity allows them to react to disturbances in their environment, leading to observable changes in behavior, such as increased activity or migration away from

Table 5
Biological indicators used in TLK for predicting and forecasting earthquakes.

Amphibian	Frogs behaving unusually [Indonesia] [87].
Annelida	Worms behaving unusually [Indonesia] [87].
Arthropoda	Ants and other insects behaving unusually [Indonesia] [87]. When insects like crickets intermittently stop making a sound for two-three minutes during the night [Philippines] [88].
Mammal	Unusual behaviors in animals, such as cats, dogs, cows, mice, elephants, horses, and pigs, have been observed in Indonesia as indicators of earthquakes [53]. In China, dogs have been reported to bark frequently and restlessly while looking at the sky, crawl on the ground, and howl [89]. Additionally, bats flying away abruptly in Indonesia have been noted as a potential sign of an impending earthquake [90].
Pisces	Fish behaving unusually [Indonesia] [87]. Fish floundering on the beach [Indonesia] [98].
Reptiles	Lizards behaving unusually [Indonesia] [87]. Snake coming out of its hole [Indonesia] [90].
Birds	Huge number of White herons flocking back to mainland during day time [Indonesia] [90].Huge number of White herons flocking back to mainland during day time [Indonesia] [90].

affected areas (Table 5).

Mammals, such as dogs, cats, cows, mice, and larger animals like elephants, demonstrate restless or unusual behaviors, including altered vocalizations and movements (Table 5). Elephants, known for their strong social structures and communication skills, have been observed to exhibit signs of distress or altered movement patterns before seismic events [94]. Research indicates that elephants may be able to detect vibrations through their feet or low-frequency sounds through their sensitive hearing [95]. This ability to sense changes in their environment allows them to react and potentially move to safer areas before an earthquake occurs which is taken traditionally as a sign of impending earthquakes.

Birds are also noted for their unusual behaviors before earthquakes. Observations have documented that birds may change their vocalization patterns, feeding habits, or flocking behavior in the days leading up to a seismic event [95]. This change is hypothesized to be linked to their acute hearing and ability to detect subtle shifts in atmospheric pressure or vibrations that humans cannot perceive [96].

Reptiles, like many other animals, have been observed to exhibit behavioral changes prior to earthquakes. These changes may include increased restlessness or movement away from typical habitats [97]. The mechanisms behind these behaviors are not fully understood, but it is believed that reptiles can detect changes in environmental factors such as temperature, humidity, and even electromagnetic fields that occur before seismic events [96]. For instance, some studies suggest that reptiles may be sensitive to the release of gases from underground fractures, which could signal an impending earthquake [91].

3.5.1.2. Meteorological indicators. Meteorological indicators of earthquakes include notable anomalies in cloud formations, temperature, and water levels. Vertical, tornado-like black clouds with calm surroundings have been reported, alongside an unusual rise in temperature as indicators of potential earthquakes (Table 6). Research has shown that variations in temperature, particularly in multiple air layers, can precede seismic events [99]. For instance, anomalies in thermal readings have been observed in the atmosphere before earthquakes, suggesting that heat generated from tectonic stress can influence local weather patterns, including cloud formation [99]. Additionally, temperature changes in groundwater have been documented as precursors to earthquakes, with fluctuations often linked to tectonic stress [100,101]. These temperature variations can alter the physical properties of water, potentially leading to observable changes in local ecosystems.

Groundwater levels can also serve as indicators of seismic activity. Studies have shown that significant changes in groundwater levels often occur prior to earthquakes, particularly in areas near active faults [101,102]. These changes can be attributed to the stress exerted on geological formations, which can lead to shifts in water pressure and flow patterns. Monitoring these fluctuations can provide critical information for predicting seismic events.

3.5.2. Volcanoes

A volcano is a geological structure where magma (molten rock), gases, and ash escape from beneath the Earth's surface. Volcanoes form when molten material from the Earth's mantle rises through cracks in the crust, often creating dramatic eruptions [104]. Since volcanoes are sudden events, it is essential to understand environmental cues to reduce the casualties and losses in their proximity.

3.5.2.1. Biological indicators. The behaviors of various animal species and plants can provide insights into impending volcanic eruptions (Table 7). Our review showed similar accounts throughout world such as cluster of yellow butterflies flying away from Merapi Volcano [105], wild chickens running away from Mayon Volcano [69], birds flying away and roosters crowing at unusual hours, including unusual singing sounds from certain indigenous birds [106], wild pigs running away from Mayon Volcano [69], extended silence from wildlife or domestic animals [106] and falling leaves of fig trees [106].

Arthropods may display increased activity or migration patterns as they sense changes in soil gases or temperature fluctuations [107]. Birds have been observed to alter their vocalizations and feeding habits, potentially in response to changes in atmospheric pressure or gas emissions from volcanic activity [107]. Mammals, particularly larger species like elephants, have shown sensitivity to ground vibrations and changes in the environment that may precede eruptions. Elephants, for example, are known to detect low-frequency sounds and vibrations, which can indicate geological disturbances [107]. Their social structures allow them to communicate these changes, potentially leading to group movements away from danger. Many animals possess heightened sensory capabilities that allow them to detect subtle changes in their environment. For example, animals can sense variations in gas concentrations, such as sulfur dioxide, which is often released prior to volcanic eruptions [107]. Additionally, changes in electromagnetic fields and vibrations can be detected by certain species, enabling them to respond to geological changes before they become apparent to humans.

Plants may respond to changes in soil moisture and nutrient availability, which can be affected by volcanic activity. The release of gases and ash can alter the chemical composition of the soil, prompting changes in plant growth patterns and community dynamics [10,108].

Table 6
Meteorological indicators used in TLK for predicting and forecasting earthquakes.

Cloud	Appearance of vertical tornado like black cloud with no wind and calm nature [Indonesia] [53].
Temperature	Rise in temperature [Indonesia] [53].
Water	Increase in water level of well [China] [89]. Receding level of sea water [Indonesia and Solomon Island] [98,103].

Table 7
Biological indicators used in TLK for predicting and forecasting earthquakes.

Arthropoda	Cluster of yellow butterflies flying away from Merapi Volcano [Indonesia] [105].
Birds	Wild chickens running away from Mayon volcano [Philippines] [69]. Birds flying away. Roosters crowing at unusual hours. Singing sound from certain indigenous bird [Papua New Guinea] [106].
Mammals	Wild pig running away from Mayon volcano [Philippines] [69]. No noise from wildlife or domestic animals i.e. extended silence [Papua New Guinea] [106].
Plants	Falling of leaves of fig tree [Papua New Guinea] [106]. Grasses dying around the top of the volcano [Papua New Guinea] [106].

3.5.2.2. Meteorological indicators. Meteorological and geological indicators of volcanic activity include distinctive environmental changes such as the presence of a blue smoke ring around a volcano and prolonged low tides (Table 8). Thunder and lightning, particularly originating from the direction of the volcano, have also been reported. One of the most significant indicators of an impending eruption is increased seismic activity. Earthquakes often occur as magma moves through the crust, creating pressure and fracturing surrounding rocks. For instance, during the 2006 eruption of Augustine Volcano, over 3,500 earthquakes were recorded in a month leading up to the eruption, demonstrating a clear correlation between seismic events and volcanic activity [109]. The frequency and intensity of these seismic events can be quantitatively analyzed using tools like the frequency index, which assesses the ratio of high-to low-frequency energy in seismic data. This method has proven effective in retrospectively identifying eruption precursors.

The release of volcanic gases, particularly sulfur dioxide (SO_2), serves as another critical indicator. Increased SO_2 emissions can signify rising magma and potential eruptions. Research indicates that monitoring SO_2 levels using instruments like the Ozone Monitoring Instrument (OMI) can enhance the detection of volcanic eruptions, as it allows for the assessment of total gas loading rather than just column amounts [110]. The presence of gas emissions can also affect atmospheric conditions, leading to climate impacts that further complicate eruption predictions [111].

Ground deformation is a crucial geological indicator of volcanic activity. Techniques such as InSAR (Interferometric Synthetic Aperture Radar) have been employed to monitor changes in the Earth's surface, which can indicate magma accumulation beneath a volcano [112]. For example, the Piton de la Fournaise volcano has been effectively monitored using PSInSAR, which allows for the detection of pre-eruptive ground deformation, providing valuable data for eruption forecasting. Such deformation often occurs as magma intrudes into the crust, causing the ground to swell. Changes in surface temperature can also indicate volcanic unrest. Increased thermal activity can be detected through remote sensing technologies, which can identify hotspots associated with rising magma [113]. The integration of multisensor data and artificial intelligence has been proposed to enhance the monitoring of these thermal anomalies, potentially leading to better eruption forecasting.

The movement of magma generates pressure that leads to fracturing and seismic activity [114]. The release of volcanic gases is a direct result of magma ascent, which reduces pressure and allows gases to escape [110]. Magma accumulation causes the surrounding rock to deform, which can be measured and analyzed to predict eruptions [112].

3.5.3. Flood

A flood occurs when water inundates land that is normally dry, typically as a result of excessive rainfall, river overflow, melting snow, storm surges, or failure of an impoundment such as an ice dam. Floods are among the most common and devastating of natural disasters, causing loss of life, destruction of property, and environmental damage. Flood prediction relies heavily on key indicators spanning astronomical, biological, and meteorological cues.

3.5.3.1. Astronomical indicators. Astronomical indicators of floods include various observations of celestial phenomena (Table 9) such as a meager figure (thin, faint, or dim) of the moon and its changing during October to November [74], formation of circular halo around the moon from August to January [115] and observation of Orion constellation from October to January [Malawi] [115].

Such prediction of floods through celestial phenomena such as the moon, stars, and sky involve understanding the gravitational influences and atmospheric conditions that contribute to hydrological events. The moon's gravitational pull is a significant factor in tidal movements, which can lead to tidal flooding, particularly during seasonal high tides. This phenomenon is characterized by periodic and predictable tidal patterns, which are exacerbated by minor wind events, leading to increased flooding frequency in coastal areas [116]. The gravitational effects of the moon and sun create a tidal cycle that can influence river levels and, consequently, flood risks in adjacent areas. The moon's appearance plays a significant role, with larger halos suggesting abundant rainfall and smaller halos indicating minimal precipitation (Table 9).

Stars have historically played a role in traditional flood prediction methods. Observing the night sky was often used to anticipate seasonal changes, which could correlate with rainfall patterns and subsequent flooding events. While modern meteorological techniques have largely supplemented these methods, the underlying principles of celestial influence on weather patterns remain relevant. For instance, large-scale atmospheric circulation patterns, which can be influenced by solar activity, have been shown to affect regional flood risks [117].

Table 8
Meteorological and geological indicators used in TLK for predicting and forecasting earthquakes.

Geological	Presence of a blue smoke ring around the volcano in April time [Papua New Guinea] [106].
Thunder and lightning	Lightning from volcanic direction [Indonesia] [105].
Water	Prolonged low tide [Papua New Guinea] [106].

Table 9
Astronomical indicators used in TLK for predicting and forecasting flood.

Moon	Bigger encirclement around the moon indicates sufficient rain and a meager figure indicates a sign of starvation and little rainfall. Changing of direction of the moon during October to November [Zimbabwe] [74]. Formation of circular halo around the moon from August to January [Malawi] [115]. Approaching full moon during the peak of rainfall [Nigeria] [118].
Sky	Black sky above the headwaters area [Indonesia] [90].
Star	Observation of orion star from October to January [Malawi] [115]. Based on purwa constellation on traditional calendar [India] [119].

3.5.3.2. Biological indicators. Biological indicators of floods include behavioral changes in amphibians, arthropods, birds, mammals, and reptiles, as well as shifts in plant morphology and phenology (Table 10). The ability of various organisms, including amphibians, arthropods, birds, mammals, reptiles, and plants, to sense impending floods combines behavioral ecology, evolutionary biology, and environmental science.

Our review documented croaking of frogs [74,118,119], Hylaranas and other frogs building nests high up in trees; toads roaring at night continuously in July–September [120] as indicators of floods. Amphibians, particularly those inhabiting flood-prone areas, have evolved behaviors that allow them to sense changes in their environment that precede flooding. For instance, they are highly sensitive to changes in humidity and atmospheric pressure, which can indicate impending rain and subsequent flooding. Studies have shown that amphibians often migrate to higher ground in anticipation of floods, utilizing their acute sensory systems to detect these environmental cues [121]. The reproductive cycles of certain reptiles are also closely tied to seasonal flooding, with nesting behaviors adapted to avoid inundation [122].

Similarly, disappearance of termites, shining fireflies (mwini-mwini) in huge numbers at night [74], presence of butterflies [123], ants moving eggs to higher ground as river levels rise [90,120], small red ants traveling with eggs and black ants with eggs moving in a straight line [119] were some of the indicators of floods used by local communities. Arthropods, including insects and spiders, exhibit a range of adaptations to cope with flooding. Many terrestrial arthropods possess behavioral strategies such as climbing vegetation or seeking refuge in elevated areas as floodwaters rise [124].

Animals are used proxy of predicting floods. Our review showed that sacred horses and sheep near lakes showing unusual behavior [125], wild pigs collecting leaves and twigs to build "mini houses" [88]; field rats moving dens to higher ground [90], migration of hippos; observation of mice in fields [115], a dog jumping into a drain to swim [126] were considered indicators of impending floods (Table 10). Birds, particularly those that inhabit wetlands, are adept at sensing environmental changes that signal flooding. They often rely on visual cues, such as rising water levels and changes in vegetation, to anticipate floods. Some species have been observed to alter their foraging and nesting behaviors in response to flooding events, moving to higher ground or shifting their nesting sites to avoid inundation. Unusual sounds from birds [74,127]; groups of big black phanga birds flying fast towards mountains [123] are some of the

Table 10
Biological indicators used in TLK for predicting and forecasting flood.

Amphibians	Croaking of frogs [Zimbabwe, Malawi, and Nigeria] [74,118,123]. Croaking of yellow-bellied frog. Tremendous noises from frog 'bengchi' [India] [119]. Hylaranas and other frogs bulding nests high up in trees. Toads roaring at night and grinding teeth continuously in the month of July, August, and September [Vietnam] [128].
Arthropoda	Disappearance of termites. Shining of fire flies (mwini-mwini) during the night in huge numbers [Zimbabwe] [74]. Presence of butterflies [Malawi] [123]. Ant moving egg to a higher ground when the river level becomes higher. Ant moving to higher place and spreading on the ground [Indonesia] [90]. Small red ants traveling with their eggs in their mouth. Special black ants with eggs moving in a straight line [India] [119]. Ant movements [India and Malawi] [119,123]. Ants building nests high up in tree. Mason bees building low nests indicates arrival of minor flood whereas building high up nest signifies sever flood [Vietnam] [128].
Birds	Unique and unusual sound from birds [Zimbabwe] [74,127]. Groups of big black phanga birds flying fast towards the mountains [Malawi] [123]. Birds producing a specific sounds, e.g. trumpet bird producing 'n'gombe n'gombe' sound in December [Malawi] [115]. Hens spreading their feathers under the balcony of house [Nepal] [61]. The disappearance or migration of certain birds from island, especially Kite and Swallow indicate approaching flood, whereas their appearance shows that the flood is about to stop [Nigeria] [118].
Mammals	If sacred horses and sheep that live at lakes in the headwater areas of Gongpa-Rangchong Valley are offended [India] [125]. When wild pigs start collecting leaves and twigs to form or build a "mini house" in a relatively flat and safe spot in the mountains, a typhoon or heavy rainfall that may lead to flooding and landslide is about to happen [Philippines] [88]. Field rats moving their dens to a higher place [Indonesia] [90]. Migration of hippos from the rivers to fields. Observation of Mice in field [Malawi] [115]. Presence of hippopotamus or its fresh fecal matter in the inland [Ghana] [50]. A dog jumping into drain to swim [Taiwan] [126].
Reptiles	Migration of crocodiles from the rivers to fields [Malawi] [115].
Plant Morphology	Change in vegetation color i.e. turning deep green [Malawi] [123]. Deep nodes on the leaves of reed grass indicate severe flood and shallow nodes indicate minor flood. Number of nodes across the leaves of bermuda grass and torpedo grass corresponds to the number of floods in a year. Red flowers on torpedo grass indicates no rain whereas white flowers indicates rain [Vietnam] [128].
Plant Phenology	Flowers of tamarind tree ('bwemba') from August to December. Increase in production of fruits of the mango tree from July to December. Plenty of bamboo ('bende') growing next to river banks from November to January [Malawi] [115]. Growth of new leaves in tree [Zimbabwe] [127]. Blooming of flowers in plume grass indicates the end of floods. If the flower blooms early in September or October in the lunar calendar, then there will be no flood. Jackfruits growing high on trees predict high flooding and growing low predicts that the flood will reach that level of fruit. Lots of bamboo flower blooming anticipate severe floods [Vietnam] [128]. Leaves growing faster and more densely [Taiwan] [126].
Month/year	In March to April and sometimes in January [Malawi] [123]. Around January and February [Zimbabwe] [129]. In the third week of April of the Lunar calendar [Vietnam] [130].
Insects	Too much of pests such as kampekete at the end of November after germination [Zimbabwe] [74]. Increased in the number of ants i.e. 'nyerere' in the villages from July to December [Malawi] [115]. Red ants with light transparent wings flying in large number [India] [119].
Pisces	Increased number of specific fish species [Malawi] [115].

potential cues observed by people. Many species can detect changes in water levels and temperature, prompting them to seek refuge in burrows or on higher ground.

Plants have been mostly used to observe their response to flood rather than as predictors of floods. Potential cues include vegetation turning deep green [123], deep nodes on reed grass leaves [128], tamarind trees flowering (August–December), mango trees increasing fruit production (July–December), bamboo growing near riverbanks (November–January) [115], new leaves growing on trees [127], and jackfruits growing higher up [128]. The flowering of lume grass marks the end of floods.

3.5.3.3. Meteorological indicators. Flooding is a complex hydrological phenomenon influenced by various meteorological indicators. Several key meteorological features (clouds, temperature, thunder and lighting, water, winds and rainbows) signal the potential for flooding (Table 11). Our review documented that meteorological indicators of floods include black and slow-moving convective clouds [123], dark clouds [127], clouds from the northwest [119], rise in temperature, localized storm events with thunder and black clouds [123], lightning in the north/northeast [61], change in river water color and smell [118,131], rise in steam water [125], increased water velocity, debris, and changes in water color (brown, black, milky, red) [123], large quantities of silt, vegetation, and rubbish in water [132], dirty and muddy water from October to January, louder river sounds from December to March, change in wind direction [115,123], prolonged heavy rainfall [89,115], double rainbows [128], and frequent appearances of rainbows [118].

Studies have shown that increased frequency and intensity of extreme precipitation events are directly correlated with climate change, leading to a higher likelihood of flooding in many regions [133–135]. Additionally, the spatial and temporal distribution of rainfall plays a critical role in flood generation, as localized heavy rainfall can overwhelm drainage systems and lead to flash floods [136]. Another significant meteorological factor is the interaction between storm surges and river discharge, particularly in coastal areas. The compounding effects of storm surges and heavy rainfall can exacerbate flooding risks, especially in delta regions where riverine and oceanic influences converge [137,138]. Moreover, changes in atmospheric conditions, such as rising temperatures and altered weather patterns, contribute to shifts in flood generation processes. For example, warmer temperatures can lead to increased evaporation and subsequently more intense rainfall events, which are critical for flood prediction [133,134]. Additionally, the alteration of land use and cover due to urbanization can exacerbate flood risks by increasing impervious surfaces, thereby enhancing runoff during precipitation events ([139,140]. This interplay between meteorological conditions and human activities underscores the complexity of flood dynamics and the necessity for comprehensive flood risk management strategies.

Thunderstorms, especially those associated with black clouds, are commonly linked to localized floods, while the direction of lightning can suggest strong winds and potential flood conditions. Changes in water conditions, such as altered river or stream water color, increased debris, and rising water levels, are key indicators of impending floods. Sudden shifts in water velocity, volume, or even smell can also signal significant flooding risks. Additionally, shifts in wind direction or the occurrence of strong winds, especially towards the end of the wet season or preceding storms, are often associated with flood events. Extended periods of rainfall, particularly after droughts, or long-duration rain events, are clear signs of possible floods. Finally, the appearance of rainbows, with overlapping rainbows indicating more severe flooding, serves as another visual cue for flood prediction. Together, these meteorological factors provide a comprehensive understanding of flood risks and help in their prediction.

Table 11
Meteorological indicators used in TLK for predicting and forecasting flood.

Cloud	Black and slow-moving convective clouds [Malawi] [123]. Presence of dark clouds i.e. amayezi amnyama [Zimbabwe] [127]. Clouds appearing from northwest corner of the sky brings rain whereas, clouds from southern corner never brings rain [India] [119].
Temperature	Rise in temperature [Malawi] [123]. Hot temperature from August to December [Malawi] [115].
Thunder and lightning	Highly localized storm events and thunder. Thunder combined with black cloud [Malawi] [123]. Lightning in the north/northeast direction means strong wind and floods [Bangladesh] [141].
Water	Change in color of the river water. Sulfuric smell in river water [Iceland] [131]. Sudden rise in the level of the stream water [India] [125]. Higher soil water saturation in the flat plain along Lake Malawi in April. Increase in water velocity and volume and amount of debris carried along with the change in color of water i.e. brown, black, milky, or red [Malawi] [123]. Large quantities of slit, vegetation debris, and rubbish present in flowing water [Cambodia] [132]. Ripe nuts floating on the river indicates heavy rain taking place in a mountain area that most likely bring large amount of water downstream [Indonesia] [90]. Colour of water getting dirty and muddy from October to January. Increase in sound of water in the rivers from December to March [Malawi] [115]. Rising level of river water [Malawi, Nepal, and India] [115,119,142]. Strange sounds from river/torrents and muddy smell in the water [Nepal] [142]. Presence of underground water while cultivating the crops [Uganda] [38]. Gerua (ochre) or peela (yellow) colour of water in river [India] [119]. Offensive smell such as rotten eggs from the river. The excessive foaming and bellowing sound from the river. The swelling of the river or increase in the river eddies [Nigeria] [118].
Wind	Change in wind direction. Strong south-easterly winds at the end of the wet season [Malawi] [123]. Mamanwas believe that Kanaway i.e. windy and stormy weather intensifies when it meet with Kabunghan i.e. sunny and windy weather which may cause severe flooding in the month of September to December up until January [Philippines] [88]. Strong winds i.e. mwera from September to January [Malawi] [115]. Strong dense winds blowing from Indian ocean towards Swaziland i.e. from east to west [South Africa] [53]. Southwest wind. Wind from the southeast or east i.e.locally known as Eshan Kun or pubal wind [Bangladesh] [141]. Heavy wind blowing from the river towards the south direction. Heavy wind approaching in May [Nigeria] [118].
Rain	Rainfall after long period of drought brings strong flood but the flood water won't last for long period [Cambodia] [132]. Heavy and long duration of rain [China and Malawi] [89,115].
Rainbow	Appearance of one rainbow means minor flood whereas two rainbows overlapping eachother means severe floods [Vietnam] [128]. Frequent appearance of rainbow indicate that the flood of that year would be of great magnitude [Nigeria] [118].

3.5.3.4. Other indicators. In Indonesia, villagers report hearing a bobbing sound in the mountains, followed by a rumbling in the sea, as an indicator of an impending flood. Traditional dating methods, such as *kutika* or *keununongis*, used in Meunasah Ray village, employ formulas like *keununong sa* and *keununong dua ploh lhee* to predict flood occurrences [90]. In Malawi, elderly community members often experience pain in specific body parts, while villagers report difficulty sleeping due to increased temperatures between September and December, both of which are considered signs of an upcoming flood [115].

3.5.4. Mass movement

Mass movements, also known as landslides or mass wasting, refer to the downward movement of rock, soil, and debris under the influence of gravity. These movements can occur rapidly or slowly and are typically triggered by factors like rainfall, earthquakes, volcanic activity, or human activities [143]. Mass movements can cause significant damage to infrastructure, ecosystems, and human life [144].

Mass movements are predicted by people by observing the behavior of certain animals. Mammals, such as sheep, exhibit unusual behavior before a landslide, often acting restless or agitated, which may signal seismic activity or shifting soil [131] (Table 12). In primates, similar behavioral changes have been noted, such as orangutans making pounding sounds within the forest, potentially in response to vibrations or disturbances in their environment [90]. These behavioral shifts may serve as early warning signs, reflecting the animals' sensitivity to changes that precede a landslide.

3.5.4.1. Meteorological indicators. Meteorological indicators such as rainfall and wind, along with pedological characteristics, play a crucial role in predicting mass movements, including landslides (Table 13). The relationship between these factors is multifaceted, involving the interplay of climatic conditions, geological characteristics, and soil properties.

Rainfall is a primary meteorological factor influencing landslide susceptibility. Heavy precipitation events can saturate soils, increasing pore water pressure and reducing the effective stress on slopes, which can lead to failure. For instance, studies have shown that extreme rainfall events are significantly correlated with landslide occurrences, particularly in regions with steep topography and loose soil structures [145]. The accumulation of water in cracks and fissures can exacerbate this risk by increasing the load on underlying materials, making them more susceptible to sliding [146]. Furthermore, the intensity and duration of rainfall are critical in establishing thresholds for landslide initiation, with empirical models often used to predict these thresholds based on historical data [145,147].

Wind, while less directly associated with landslides than rainfall, can influence soil moisture levels and erosion processes. In some contexts, strong winds can contribute to the drying of soils, which may lead to increased susceptibility to erosion and subsequent landslides when rainfall does occur [148]. Additionally, wind can affect the microclimate of an area, potentially altering vegetation cover and soil stability over time [148].

Pedological characteristics, including soil type and weathering processes, are also vital in understanding mass movement phenomena. Different soil types exhibit varying degrees of cohesion and shear strength, which directly impacts slope stability. For instance, weathered rocks and soils that have undergone significant alteration due to weathering processes are often more prone to landslides due to their reduced structural integrity [149,150]. The presence of clay minerals, which can swell and shrink with moisture changes, further complicates this relationship, as they can lead to increased instability during wet conditions [151].

Moreover, geological factors such as lithology and tectonic activity can influence both the susceptibility to landslides and the effectiveness of meteorological indicators in predicting them. Areas with certain rock types may be more prone to erosion and mass movement, particularly when combined with heavy rainfall and seismic activity [149,152] interaction between these geological characteristics and meteorological conditions creates a complex framework for assessing landslide risk.

3.6. Meteorological disasters

Meteorological disasters are extreme weather events that cause significant damage to life, property, and the environment. These disasters are often unpredictable and can vary in severity, but they are primarily driven by atmospheric conditions such as temperature, pressure, moisture, and wind [156]. Common meteorological disasters include hurricanes, cyclones, tornadoes, blizzards, droughts, and floods.

3.6.1. Storm

A storm is a disturbance in the atmosphere characterized by strong winds, precipitation (rain, snow, hail, etc.), thunder, lightning, or other weather phenomena [157]. Storms can vary greatly in size, intensity, and type, and can range from localized thunderstorms to massive cyclones or hurricanes. Depending on the severity, storms can cause significant damage to infrastructure, ecosystems, and human lives.

3.6.1.1. Astronomical indicators. Astronomical indicators such as the sun, moon, stars, and the overall appearance of the sky can provide valuable insights into impending storms (Table 14). These celestial bodies and phenomena exhibit specific characteristics that,

Table 12
Biological indicators used in TLK for predicting and forecasting mass movements.

Mammals	Unusual behavior of sheep [Iceland] [131].Pounding sound of orangutan echoed within the forest [Indonesia] [90].

Table 13
Meteorological indicators used in TLK for predicting and forecasting mass movements.

Rain	Heavy and long duration of rainfall [China, Indonesia, and Srilanka] [15,89,153–155]
Season	Spring season [Indonesia] [155].
Wind	Mamanwas believe that Kanaway i.e. windy and stormy weather intensifies when it meets with Kabunghan i.e. sunny and windy weather that may cause severe landslides in the month of September to December, until January [Philippines] [88].
General	Unusual earth cracks [Sri Lanka, Indonesia and China] [15,89,153,154]. Ground vibration and ground crack [Indonesia] [155]. The water oozing up from underground combined with severe soil erosion from down the slope [Uganda] [38].

Table 14
Astronomical indicators used in TLK for predicting and forecasting storms.

Sky	Dark blue sky [Fiji] [122]. Still skies [Australia] [158]. Drizzling and gloomy sky [Bangladesh] [159]. Darkening of sky with apperance of abuabu (cloudlike fog) on the horizon [Philippines] [69]. Change in colour of sky [Philippines] [160].
Star	Position of star [Nepal] [142]. Absence of the stars [Philippines] [160]. Visibility of constellations of stars [Indonesia and Philippines] [16].
Moon	Formation of ring around moon; nearer the ring around the moon, closer the typhoon [Philippines] [161]. Large ring around the moon [Philippines] [160].
Sun	Observation of colour of the horizon during sunset [Philippines] [160]. Formation of ring around sun; nearer the ring around the sun, closer the typhoon [Philippines] [161].

when observed, can signal changes in weather patterns, often before meteorological instruments detect them. Our review found a variety of indicators such as dark blue sky [69,122], still skies [158], drizzling and gloomy sky [159], position of stars [142], absence of stars [160], visibility of star constellations [16], formation of a ring around the moon where a smaller ring indicates that a typhoon is nearer [160,161], observation of horizon color during sunset [160] and formation of a ring around the sun [161].

The position of sun, moon, star and their other attributes are closely linked to seasonal and annual weather patterns and may signal a storm through variety of pathways (Table 14). The overall appearance of the sky, including cloud formations and colors, can provide

Table 15
Biological indicators used in TLK for predicting and forecasting storms.

Annelids	Balolo i.e. sea worms becoming dirty [Fiji] [122,166]
Arthropods	Hornets moving their hives to lower level [Fiji] [122,166]. Ants staying together and climbing to higher places. Insects flying in a rush and less visible at night. Mosquitoes sticking in the body of cows and goats [Bangladesh] [141]. Hornets nesting close to the ground [Fiji] [167]. Bees relocating their hive to higher ground or branches higher above ground level [Fiji]. Ants collecting food and migrating to higher locations [Philippines]. Ants climbing toward roof of houses. Cluster of bees or locusts in the sky. Insects or flies biting the cattle [Bangladesh] [159]. Mobility of ants [Nepal] [142].
Birds	Unusual sound and movement of birds [Fiji and Nepal] [122,142,166]. No bird flying in the sky [Australia] [158]. Guyalla birds flying out to sea. Livestock such as ducks and hens becoming reluctant to enter their shelters. Seabirds and ducks starting to move toward the shore. Small black birds start flying over the sea [Bangladesh] [141]. Seabirds flying landward [Fiji] [141]. Birds native to smaller island groups flying over the mainland i.e. Mamanuca island group [Fiji] [172]. Abnormal birds flying without destination. Sea birds and pigeons moving toward inland [Bangladesh] [159]. Suddenly stop of chirping of certain bird species like rufous hornbill, Palawan hornbill, and white-eared brown dove during the daytime [Philippines] [88]. Domesticated ducks flying to other areas [Philippines] [69]. Chickens moving to the roof of the house [Philipines] [160]. Kiling bird (Luscinia calliope) makes sound to indicate the passing of last typhoon. And, its throaty sound means still another typhoon is coming. Presence of bird tiway in august or 1st week of september [Philippines] [161]. Flocks of birds flying to the mountains [Vietnam] [128].
Crustaceans	Crabs coming into courtyards or high place [Bangladesh] [159]. Crabs coming out of the ground and scampering to other areas. Unusual appearance of crab (talangka) in the surface of the ocean [Philippines] [69] Hermit crabs going inland or climbing up trees all forewarn [Philippines] [16]. Crabs moving to the dry land [Philippines] [161].
Chondrichthyens	Unusual appearance of tame shark (mabait na pating) in the surface of the ocean [Philippines] [69].
Echinoderms	Sea urchins and starfish appear in the shorelines [Philippines] [69].
Mammals	Change in dog behavior [Australia] [158]. Cows start starvation [Bangladesh] [141]. Dogs hiding [Fiji] [167]. Cattle and dogs being very vocal in the day time [Fiji] [172]. Cattle and dogs howl endlessly at the night [Bangladesh] [159]. Rats coming out of the ground and scampering to other areas [Philippines] [69].
Pisces	Churi (*Eupleurogrammus muticus*) and Loittia fish (*Harpadon nehereus*) moving and jumping quickly in the sea [Bangladesh] [141]. Jumping of fish in ponds [Bangladesh] [159]. Absence of fishes in shallow waters. Sudden surfacing of tongue-fish and rabbit fish in abundance [Philippines] [69]. Significantly reduced number of barracuda i.e. a popular catch fish [Fiji] [172].
Amphibian Behavior	Frog emerging from water [Philippines] [69].
Reptiles Behavior	Fast movement of sea snakes [Philippines] [16]. Snakes coming out in the ground and scampering to other areas. Sudden surfacing of sea snake in abundance. Unusual appearance of sea snake (walowalo) in the surface of ocean [Phlippines] [69].
Plant Morphology	Leaves of Mantha tree getting curled [Bangladesh] [141]. Vudi or plantain's shoot bending downward [Fiji] [122,166]. Smaller size of tomatoes (Solanum lycopersicum) that can fall from the vine earlier than usual [Fiji] [172]. Inverted leaves of star apple trees (caimito). Seaweed seen in an upright position [Philippines] [69].
Plant Phenology	New growth on banana plants [Fiji] [122]. Sudden growth of bamboo shoots [Fiji] [122,166]. More than one breadfruit on one stalk [Fiji] [167]. Fruits of mangoes and breadfruit in high abundance [Fiji] [172]. Mango or tamarind growing off season. Reeds, flowers, or fruits (particularly breadfruits) growing in high abundance [Fiji] [166]. Golden shower plant starts to bloom during summer [Philippines] [160]. Blooming of flower in plume grass denotes end of storm [Vietnam] [128].
Plant Physiology	Mango or tamarind growing off season [Fiji] [166].

immediate clues about impending storms. For example, the presence of dark, towering cumulonimbus clouds is a strong indicator of severe weather, including thunderstorms and heavy rainfall [162]. Additionally, phenomena such as halos around the sun or moon can indicate moisture in the atmosphere, suggesting that a storm may be approaching [163].

The sun's position and intensity can indicate changes in weather. For instance, a sudden decrease in sunlight intensity, often accompanied by a rapid increase in cloud cover, can suggest that a storm is approaching. This phenomenon is primarily due to the formation of thick clouds that block sunlight, which is a precursor to precipitation [162]. Additionally, the sun's position in the sky can influence local temperature and humidity levels, which are critical factors in storm development. For example, a significant drop in temperature during the day may indicate the arrival of a cold front, which can lead to storm formation.

The moon plays a crucial role in weather prediction, particularly through its influence on tides. The gravitational pull of the moon affects ocean tides, which can interact with storm surges to exacerbate flooding during severe weather events [164]. Observations of the moon's appearance, such as its color and size, can provide insights into upcoming weather changes, including storms [163]. For example, a full moon may correlate with increased storm activity, as seen in studies of seabird fledging patterns that align with lunar cycles [165].

The positions and movements of stars have historically been used for weather forecasting including storms. Certain constellations and star formations can indicate seasonal changes that correlate with storm patterns. For instance, the appearance of specific stars in the evening sky can signal the transition into a storm-prone season [163]. This practice is rooted in the understanding that astronomical cycles are closely linked to climatic changes, as observed in various indigenous cultures [162]. The visibility of stars can also be affected by atmospheric conditions: a clear night sky typically indicates stable weather, while a hazy or obscured sky may suggest moisture in the atmosphere, which can precede storms.

3.6.1.2. Biological indicators. Biological indicators of impending storms encompass a variety of organisms, including animals, birds, plants, insects, and annelids. These organisms exhibit behavioral and physiological changes that signal the approach of severe weather, often before humans can perceive such changes.

Notable indicators documented by our review related to annelids and insects include balolo (sea worms) becoming dirty [122,166], hornets nesting closer to the ground or at lower levels [166,167], ants climbing to higher places, migrating, collecting food, and clustering on roofs of houses [141,142,159] and insects flying in a rush, less visible at night, biting cattle, or clustering [159] (Table 15). Annelids' response may be related to soil moisture and other environmental factors that are influenced by approaching storms. Insects also serve as important biological indicators, often altering behavior in response to storms. Arthropods like ants climb to higher places, collect food, and migrate, while hornets and bees adjust nest locations—hornets move lower, and bees relocate higher.

The movements and sounds of birds have been considered important indicators of disasters such as birds flying inland or making unusual sounds [122,128,142], ducks and hens being unwilling to enter shelters; chickens moving to roofs [69,160] and cessation of chirping by certain species during daytime [88]. Birds, particularly migratory species, have been documented to exhibit storm avoidance behaviors. Research shows that breeding songbirds, such as the golden-winged warbler, can undertake facultative migrations in response to environmental cues associated with storms, indicating their sensitivity to atmospheric changes [168]. This ability to sense shifts in barometric pressure and wind patterns allows them to vacate areas before severe weather strikes.

In aquatic environments, changes in water conditions, such as increased sedimentation and alterations in water quality, can also serve as indicators of approaching storms. Studies have documented that marine species, including sharks and dolphins, exhibit changes in movement patterns and habitat use in response to these environmental cues [169,170] as documented in our review that the unusual presence of docile sharks near the ocean surface indicates storms [69] (Table 15). Sharks have been shown to alter their spatial patterns in anticipation of severe tropical storms, likely as a means to avoid the adverse conditions associated with such events [169]. Additionally, bottlenose dolphins may adjust their foraging behavior in response to the disorientation of prey caused by storm conditions [170].

Moreover, the behavior of fish and other aquatic organisms can be influenced by the physical disturbances associated with storms [171], which is similar to reported indicators of storms such as rapid movement or jumping of fish in water bodies, reduced their visibility in shallow waters, or sudden abundance in specific species [69,159].

Plant morphology and phenology are important indicators of impending storms. The review showed storms are anticipated by tree leaves curling or bending downward (e.g., Mantha tree, plantain shoots) [141,166], smaller tomatoes falling early, inverted leaves of star apple trees [69,172], sudden growth of bamboo shoots, breadfruit or mango growing off-season [122] and high yields of mangoes, tamarind, or breadfruit [172]. Such unusual growth patterns and phenological change may are caused by ecological disturbances. The physiological stress experienced by plants during storm conditions can lead to observable changes in their growth patterns and community dynamics.

Behavioral changes in marine and terrestrial creatures before storms result from environmental cues and physiological responses. Movements like crustaceans heading inland, unusual fish activity, and altered mammal behavior reflect their ability to sense pre-storm changes, often linked to drops in barometric pressure. Many species detect these drops 12–24 h before storms [173], prompting protective behaviors. For example, Ferruginous Hawks increase nest maintenance as pressure falls, while social animals engage in group responses that enhance survival [173].

3.6.1.3. Meteorological indicators. Meteorological elements such as water levels, wind patterns, cloud formations, temperature shifts, rainfall, and phenomena like rainbows play an essential role in signaling the likelihood of storms (Table 16). Fluctuations in water

Table 16
Meteorological indicators used in TLK for predicting and forecasting storms.

Cloud	No cloud in the dark blue sky [Fiji] [122]. Cloud moving very fast [Fiji] [167]. Position of cloud in the sky [Nepal] [142]. Red cloud [Taiwan] [126]. Clouds appear similar like shape of ground starched by rooster. Summit of Mayon Volcano is fully covered with clouds [Philippines] [69]. Changes in texture, color, location, and movement, including speed (fast) and direction (vertical or horizontal) of cloud [Indonesia and Philippines] [16].
Temperature	Increase in temperature [Fiji] [122,166]. Hot and humid weather [Bangladesh and Fiji],174]. Extremely hot weather [Fiji and Nepal] [133, 161]. Hot weather followed by rain [Bangladesh] [159].
Thunder and lightning	No thunder and lightning with silent (gombir) weather [Bangladesh] [141].
Water	If the tides go out, the wind goes slow but if the tides go full and high, the wind goes harder and can bring cyclone [Australia] [158]. Change in color of seawater. Elevated water levels in the adjacent rivers. Warm seawater. Whirling water and vapors in the sea surface air [Bagladesh] [141]. Drying up of nearby ponds [Fiji] [167]. Abnormally hot water in river/sea. Dark/smoky/cloudy colour of water. Gigantic waves of water in the sea. Huge roar of the sea/river. Increase of water in the river [Bangladesh] [159]. Elevated water levels in the adjacent rivers [Bangladesh] [141]. Change of sea waves to clockwise direction [Philippines] [69]. Change in colour of the river water [Philippines] [160]. High and white waves [Indonesia and Philippines] [16].
Rain	Heavy rainfall [Philippines] [16]. Extent of rainfall in the upper catchments [Nepal] [142].
Wind	Abnormal wind circulation. Muddy smell in the air. Strong wind circulation from south or south-east. Wind blowing in the deep sea as a circle [Bagladesh] [159]. Intensity of wind [Nepal] [142]. Strong winds with highwave as compared to normal days [Philippines] [69]. Northeast/southwest monsoon wind. Strong winds i.e. Alimbusamos. Wind coming from the direction where the sunrise or the sunset [Philippines] [160].
Rainbow	Halfway rainbow [Philippines] [69].

levels and changes in precipitation patterns are often among the first indications of an approaching storm. Our review identified that communities relate impending storms to a variety of indicators, including no clouds in a dark blue sky [122], increased and extremely hot weather [142,166], no thunder or lightning, silent weather [141], high tides [158] and heavy rainfall [16].

Rising humidity combined with the formation of dark clouds frequently signals impending rainfall [174]. This process involves atmospheric moisture accumulating to form towering cumulonimbus clouds, which are closely linked to thunderstorms [174]. Moreover, the intensity of precipitation can reveal the storm's severity: heavy, sustained rainfall often accompanies robust storm systems capable of causing flooding and other threats [175]. Wind patterns also serve as a vital indicator of storm activity. Increasing wind speeds or sudden changes in direction often suggest that a storm system is on the horizon. Strong gusts typically precede severe weather, reflecting the dynamics of low-pressure systems interacting with high-pressure zones. These interactions create steep pressure gradients, leading to turbulent wind conditions that can escalate into storms [176,177].

Cloud formations provide another critical clue. Cumulonimbus clouds, known for their height and density, are hallmark precursors of thunderstorms and other severe weather events. Their presence indicates atmospheric instability, a primary driver of storm activity [178]. Additionally, dramatic features such as shelf clouds or wall clouds often herald extreme conditions, including strong updrafts and even tornado potential [179].

A sudden drop in temperature is another common precursor to severe weather. This cooling effect often occurs with the arrival of a cold front, which forces warm air upwards, creating turbulence that fosters storm development [180]. In some cases, temperature inversions trap moisture near the surface, increasing the likelihood of cloud formation and eventual precipitation [181].

Even rainbows, typically associated with serene weather, can hint at changing atmospheric conditions. These colorful arcs form when sunlight refracts through raindrops, signifying localized rain while the sun shines elsewhere. Such occurrences can often signal the proximity of a storm, especially in cases of isolated thunderstorms [182]. This interplay of light, moisture, and atmospheric conditions reflects the complex dynamics leading up to severe weather.

4. Conclusions

Disaster Risk Reduction (DRR) emphasizes prevention, preparedness, mitigation, response, and recovery. Prediction and forecasting are critical components of the contemporary DRR framework, as they enable early warning systems and support preparedness through risk-informed decision-making. Traditional and local knowledge (TLK) encompasses a vast breadth of community-based understanding related to disaster prediction and forecasting. The use of TLK-based prediction helps contextualize risks, strengthen community participation in DRR, and provides a cost-effective and locally grounded option for anticipating disasters.

Understandably, there are both similarities and distinctions in the indicators—objects, situations, or both—that signal impending disasters. The temporal and spatial variability of these indicators necessitates careful interpretation and localized application. This highlights the importance of acknowledging TLK in disaster risk reduction by governments and entrusting communities with stewardship roles in disaster risk management. Therefore, the creation of appropriate policies, including supportive institutional and legal frameworks, is essential to integrate TLK effectively into DRR. However, it is equally important to document, catalogue, and validate TLK-based prediction methods to ensure their reliability and broader applicability.

We offer following additional conclusions:

1. Scientific Merit of TLK Systems

TLK systems often use astronomical, biological, and meteorological indicators to forecast natural disasters, demonstrating community connection with and understanding of environmental phenomena. The similarities in predictions across different regions, including types of disasters, species involved, and contextual factors, suggest that these observations may have scientific merit and offer valuable insights into natural patterns. Research into the scientific basis of TLK for the forecasting and prediction of disasters is limited, but of high potential for providing new perspectives on predictive approaches. This is urgent, as traditional knowledge continues to erode under rapid change in social and environmental circumstances.

2. Interdisciplinary Collaboration in Research and Implementation

TLK systems vary widely across cultures and regions, with their application being highly localized and shaped by specific microhabitats, climates, and cultural contexts. As a result, observations from one area may not be universally applicable, underscoring the importance of localized validation and contextualization. This underscores the need for interdisciplinary research involving experts from diverse fields (e. g., disaster management, ecology, and the social sciences) to bridge disciplinary gaps and enhance the integration of TLK into disaster preparedness strategies.

CRediT authorship contribution statement

Prakash Kumar Paudel: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Raja Ram Chandra Timilsina:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Formal analysis, Data curation. **Dinesh Bhusal:** Writing – review & editing, Writing – original draft, Visualization. **Henry P. Huntington:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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